

06. Open API Documentation

1. Overview

This document describes the backend APIs used by the **ICU Nursing Allocation Decision Support System**.

The system supports ICU nursing workflows including:

- shift selection
- nurse overview
- patient burden assessment
- STAT order management
- nursing task tracking
- automatic patient-bed allocation
- handoff snapshot generation
- handoff archive lookup
- war room dashboard

2. Base Information

Item	Value
Base URL	http://127.0.0.1:8787/api/v1
Data format	JSON
Authentication	Demo user based / local development
Swagger UI	Not configured in this project

Swagger UI

Currently, this project does **not** provide a Swagger UI endpoint.

If Swagger UI is added later, the URL can be documented here, for example:

```
http://127.0.0.1:8787/api-docs
```

3. Common Response Format

Most API responses follow this structure:

```
{  
  "data": {}  
}
```

```
}

```

For list responses:

```
{
  "data": []
}

```

For error responses:

```
{
  "error": {
    "code": "INTERNAL_SERVER_ERROR",
    "message": "Error message",
    "details": {}
  }
}

```

4. Common Parameters

shiftId

Many APIs require a **shiftId** because most data is shift-specific.

Example shift IDs:

Shift	shiftId
2026/05/19 白班 07:00-15:00	00000000-0000-0000-0000-000000000202
2026/05/19 小夜班 15:00-23:00	00000000-0000-0000-0000-000000000203
2026/05/19 大夜班 23:00-07:00	00000000-0000-0000-0000-000000000204

5. System APIs

GET /

Returns basic backend API information and available endpoints.

Example Response

```
{
  "data": {
    "name": "ICU Nursing Allocation API",
    "baseUrl": "/api/v1",
    "endpoints": [
      "GET /api/v1/health",

```

```
    "GET /api/v1/me",  
    "GET /api/v1/shifts"  
  ]  
}
```

GET /health

Checks whether the backend server is running.

Example Response

```
{  
  "data": {  
    "ok": true  
  }  
}
```

6. User API

GET /me

Returns the current user information.

Query Parameters

Name	Type	Required	Description
<code>userId</code>	string	No	Demo user ID

Example Request

```
GET /api/v1/me?userId=00000000-0000-0000-0000-000000000110
```

Example Response

```
{  
  "data": {  
    "id": "00000000-0000-0000-0000-000000000110",  
    "shortName": "陳0琪",  
    "role": "nurse"  
  }  
}
```

7. Shift APIs

GET `/shifts`

Returns all shifts.

Example Response

```
{
  "data": [
    {
      "id": "000000000-0000-0000-0000-0000000000203",
      "shiftKey": "evening",
      "label": "2026/05/19 小夜班 15:00-23:00",
      "startsAt": "2026-05-19T07:00:00.000Z",
      "endsAt": "2026-05-19T15:00:00.000Z",
      "status": "confirmed",
      "hidden": false,
      "chargeNurse": {
        "id": "000000000-0000-0000-0000-0000000000110",
        "shortName": "陳0琪"
      },
      "nurseIds": []
    }
  ]
}
```

GET `/shifts/current`

Returns the current or default shift.

Example Request

```
GET /api/v1/shifts/current
```

8. Nurse APIs

GET `/nurses`

Returns the nurse list.

Example Request

```
GET /api/v1/nurses
```

GET /nurses?shiftId={shiftId}

Returns nurses related to a specific shift.

Query Parameters

Name	Type	Required	Description
shiftId	string	No	Shift ID used to filter nurses

Example Request

```
GET /api/v1/nurses?shiftId=00000000-0000-0000-0000-000000000203
```

9. Nurse Overview API

GET /nurse/overview?shiftId={shiftId}

Returns overview data for the nurse homepage.

This API is used to display:

- current shift summary
- selected nurse
- assigned patients
- all patients in the shift
- workload information

Query Parameters

Name	Type	Required	Description
shiftId	string	Yes	Current shift ID

Example Request

```
GET /api/v1/nurse/overview?shiftId=00000000-0000-0000-0000-000000000203
```

10. Burden Assessment APIs

GET /burden-assessments?shiftId={shiftId}&scope={scope}

Returns burden assessment records.

Query Parameters

Name	Type	Required	Description
<code>shiftId</code>	string	Yes	Shift ID
<code>scope</code>	string	No	<code>mine</code> or <code>all</code>

Example Request

```
GET /api/v1/burden-assessments?shiftId=00000000-0000-0000-0000-000000000203&scope=mine
```

PATCH `/burden-assessments/{assessmentId}`

Updates a burden assessment.

Path Parameters

Name	Type	Description
<code>assessmentId</code>	string	Burden assessment ID

Example Request Body

```
{
  "objective": {
    "mobility": 2,
    "monitoring": 3
  },
  "subjective": {
    "isolation": true,
    "familyCommunication": "high"
  }
}
```

11. Task APIs

GET `/tasks?shiftId={shiftId}`

Returns task data for the selected shift.

Query Parameters

Name	Type	Required	Description
<code>shiftId</code>	string	Yes	Shift ID
<code>assignee</code>	string	No	Example: <code>me</code>

Name	Type	Required	Description
<code>status</code>	string	No	Example: <code>pending</code>
<code>kind</code>	string	No	Task type

Example Request

```
GET /api/v1/tasks?shiftId=00000000-0000-0000-0000-000000000203&assignee=me&status=pending
```

PATCH `/tasks/{taskId}`

Updates a task.

Path Parameters

Name	Type	Description
<code>taskId</code>	string	Task ID

Example Request Body

```
{
  "status": "completed"
}
```

12. STAT Order APIs

GET `/stat-orders?shiftId={shiftId}`

Returns STAT orders for the selected shift.

Query Parameters

Name	Type	Required	Description
<code>shiftId</code>	string	Yes	Shift ID

Example Request

```
GET /api/v1/stat-orders?shiftId=00000000-0000-0000-0000-000000000203
```

POST `/stat-orders`

Creates a new STAT order.

Example Request Body

```
{
  "shiftId": "000000000-0000-0000-0000-0000000000203",
  "admissionId": "000000000-0000-0000-0000-0000000000501",
  "title": "STAT Chest X-ray",
  "kind": "STAT",
  "urgent": true
}
```

PATCH /stat-orders/{orderId}

Updates a STAT order.

Path Parameters

Name	Type	Description
orderId	string	STAT order ID

13. Allocation Run APIs

GET /allocation-runs/current?shiftId={shiftId}

Returns the current allocation run for the selected shift.

Query Parameters

Name	Type	Required	Description
shiftId	string	Yes	Shift ID

POST /allocation-runs/suggest

Creates a suggested allocation run.

This API is used by the charge nurse to generate a suggested patient-bed allocation.

Example Request Body

```
{
  "shiftId": "000000000-0000-0000-0000-0000000000202",
  "targetShiftId": "000000000-0000-0000-0000-0000000000203"
}
```


Field Description

Field	Description
<code>shiftId</code>	Source shift used for evaluation
<code>targetShiftId</code>	Next shift that will receive the allocation result

GET `/allocation-runs/{allocationRunId}`

Returns allocation run details.

Path Parameters

Name	Type	Description
<code>allocationRunId</code>	string	Allocation run ID

GET `/allocation-runs/{allocationRunId}/items`

Returns allocation items for a specific allocation run.

POST `/allocation-runs/{allocationRunId}/confirm`

Confirms an allocation run.

After confirmation:

1. the allocation run becomes confirmed
2. the result is sealed
3. a handoff snapshot is created
4. the snapshot should be stored using the `targetShiftId`

Important Behavior

The handoff snapshot must be stored under the **target shift**, not only the source shift.

Example:

Source Shift	Target Shift	Snapshot should be stored in
2026/05/19 白班	2026/05/19 小夜班	2026/05/19 小夜班

14. War Room API

GET `/war-room?shiftId={shiftId}`

Returns dashboard data for the selected shift.

This API is used by the war room page to show:

- nurse workload
- patient distribution
- urgent task summary
- STAT order summary
- workload imbalance

Query Parameters

Name	Type	Required	Description
<code>shiftId</code>	string	Yes	Shift ID

15. Handoff Sheet API

GET `/handoff-sheets?shiftId={shiftId}`

Returns handoff sheet data for the selected shift.

Query Parameters

Name	Type	Required	Description
<code>shiftId</code>	string	Yes	Shift ID

16. Handoff Snapshot APIs

GET `/handoff-snapshots`

Returns all handoff snapshot records.

GET `/handoff-snapshots?shiftId={shiftId}`

Returns handoff snapshots for a specific shift.

Query Parameters

Name	Type	Required	Description
<code>shiftId</code>	string	Yes	Shift ID

Example Request

```
GET /api/v1/handoff-snapshots?shiftId=00000000-0000-0000-0000-000000000203
```

Example Response

```
{
  "data": [
    {
      "id": "711326f0-ea04-4f0a-9b07-c2d0b4eca7ac",
      "allocationRunId": "00000000-0000-0000-0000-000000000501",
      "shiftId": "00000000-0000-0000-0000-000000000203",
      "shiftKey": "evening",
      "shiftLabel": "2026/5/19 小夜班 15:00-23:00",
      "createdAt": "2026-06-05T15:51:29.098Z",
      "createdBy": "陳0琪",
      "summary": {
        "patientCount": 17,
        "nurseCount": 9,
        "statTotal": 33,
        "avgLoad": 39.3,
        "maxLoad": 72
      }
    }
  ]
}
```

GET `/handoff-snapshots/{snapshotId}`

Returns details of a specific handoff snapshot.

Path Parameters

Name	Type	Description
<code>snapshotId</code>	string	Handoff snapshot ID

17. Main User Flows

17.1 Nurse Flow

- 1. Select shift.
- 2. Select nurse.
- 3. View assigned patients.
- 4. Fill in burden assessment.
- 5. Check STAT orders and tasks.

17.2 Charge Nurse Allocation Flow

- 1. Select the source shift.
- 2. Enter the allocation page.
- 3. Generate suggested allocation.
- 4. Review decision trace.

5. Confirm allocation.
 6. Backend creates a handoff snapshot.
 7. The next shift can view the handoff result.
-

17.3 Handoff Snapshot Flow

```
Source shift allocation
    ↓
Confirm allocation
    ↓
Create handoff snapshot
    ↓
Save snapshot with targetShiftId
    ↓
Next shift can view the sealed handoff result
```

18. Notes

- Most APIs are shift-based and require `shiftId`.
- The frontend communicates with the backend using helper methods such as `apiGet`, `apiPost`, and `apiPatch`.
- Handoff snapshots are used to preserve confirmed allocation results.
- Confirmed allocation should save the snapshot to the target shift.
- Swagger UI is not currently configured.